

Supplementary material for "Constrained, not identifiable: preregistered tests of how the Voynich manuscript text was produced"

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Built 23 September 2026 from repository commit 922722b. All tables are generated from the result files by the same script that builds the main text.

S1. Deviations and preregistration timestamps

Table S1. Logged deviations from the preregistrations. Full texts in the repository.

File	Deviation (all made before the corresponding Voynich statistic was computed)
Battery	Gibberish formatting (data handling not foreseen in Section 4.); T6a null computed exactly.; T5 choice of K.; T1 Monte Carlo precision.; T1 null restricted to the page.; T1 estimand made page-conditional.; LATV dropped.
E1	Tuning grid extended.; Position-stratified expectation.; Inference through the steps.
E5	Uncertainty by quire jackknife instead of page bootstrap.
E2–E7	none beyond those recorded in PROGRAM.md (E6/E7: none)

Table S8. Timestamps: each preregistration was committed before the corresponding Voynich outcome was computed.

Stage	Commit	Content
Battery preregistration	28ae510	PREREG.md frozen
Battery Stage A	2403034	controls only
Battery Stage B and report	123ee76	Voynich
E1 preregistration	b5dcb79	
E1 calibration	ea64b09	before the Voynich run
E2 preregistration	6b109a4	
E3 preregistration	0f50303	
E4 preregistration	ce8bb45	
E5 preregistration	f3bb895	
E6 preregistration	537e93b	
E7 preregistration	6b6e555	
Red-team checks	922722b	post hoc

S2. Generator parameters

Table S2. Fitted generator parameters. λ (lam): quire session weight; σ (sig): page tilt; σ_w (sigw): line-to-line drift; ρ , τ , e : recency rate, range, exact share; ϵ (eps): slip rate; m : pruning threshold; θ (theta): choice temperature; c : coinage rate; sq , sp , sw : quire, page and line variation of repertoire weights.

Model	Fitted parameters (training half)
E2 HGR	lam = 0.25, sig = 0.4, rho = 0.08, tau = 8, e = 0.1, eps = 0
E3 HGR2 (fit even)	m = 1, theta = 1.0, eps = 0.08, rho = 0.05, tau = 8, e = 0.1, lam = 0.25, sig = 0.4, sigw = 0.05
E3 HGR2 (fit odd)	m = 1, theta = 1.0, eps = 0.08, rho = 0.05, tau = 16, e = 0.1, lam = 0.25, sig = 0.4, sigw = 0.05
E6 TR (fit even)	m = 1, c = 0.1, eps = 0.02, rho = 0.03, tau = 4, e = 0.1, sq = 0.4, sp = 0.4, sw = 0

S3. Held-out statistics for every generator

Table S3. E2: held-out statistics (test: even bifolios).

Statistic	Voynich	full	no recency	no pruning	no session	plain automaton
hapax share †	0.7085	0.6022 ✗	0.6314 ✗	0.6799 ✗	0.6275 ✗	0.7044 ✓
type-token ratio †	0.3012	0.2190 ✗	0.2110 ✗	0.2825 ✓	0.2261 ✗	0.2914 ✓
Zipf slope	-0.7460	-0.7448 ✓	-0.7588 ✓	-0.7537 ✓	-0.7460 ✓	-0.7633 ✓
mean word length	5.3867	5.1717 ✓	5.1360 ✓	5.2468 ✓	5.1550 ✓	5.2544 ✓
boundary MI	0.1523	0.1328 ✓	0.1645 ✓	0.1315 ✓	0.1325 ✓	0.1563 ✓
word-pair MI	0.0549	0.0255 ✓	0.0582 ✓	0.0122 ✓	0.0304 ✓	0.0340 ✓
distance-2 word MI	0.0425	0.0233 ✓	0.0192 ✓	0.0103 ✗	0.0095 ✗	0.0153 ✗
line-break word MI	0.0006	-0.0014 ✓	0.0038 ✓	0.0010 ✓	0.0040 ✓	0.0021 ✓
line-break boundary MI	0.0120	-0.0032 ✓	0.0163 ✓	0.0052 ✓	0.0016 ✓	-0.0009 ✓
boundary MI, equal n	0.1511	0.1370 ✓	0.1660 ✓	0.1246 ✓	0.1272 ✓	0.1406 ✓
space predictability	0.7388	0.7847 ✗	0.7910 ✗	0.7464 ✓	0.7772 ✗	0.7375 ✓
lexicon test, real words	0.7378	0.8133 ✗	0.8255 ✗	0.7487 ✓	0.8089 ✗	0.7481 ✓
lexicon test, synthetic words	0.7186	0.7657 ✓	0.7779 ✓	0.7183 ✓	0.7613 ✓	0.7127 ✓
conditional entropy h2	1.8787	1.8107 ✓	1.7747 ✗	1.9186 ✓	1.8226 ✓	1.9219 ✓
conditional entropy h3	1.7779	1.7097 ✓	1.6695 ✗	1.7958 ✓	1.7130 ✓	1.7954 ✓
conditional entropy h4	1.6475	1.6415 ✓	1.6066 ✓	1.6929 ✓	1.6385 ✓	1.6909 ✓
lzma bits per character	2.4440	2.3285 ✓	2.2877 ✓	2.4836 ✓	2.3491 ✓	2.4943 ✓
page MI †	0.2529	0.3048 ✓	0.2172 ✓	0.1993 ✓	0.2168 ✓	0.1731 ✗
T1 line-break ratio	0.0809	-0.1428 ✓	-0.0347 ✓	0.2131 ✓	-0.0763 ✓	-0.0125 ✓
T2a final form, margin	0.0942	0.1346 ✓	0.1441 ✓	0.1230 ✓	0.1268 ✓	0.1357 ✓
T2a final form, paragraph end	0.0373	0.1653 ✗	0.1770 ✗	0.1653 ✗	0.1691 ✗	0.1614 ✗
T2b line-fill slack	1.9622	9.4848 ✗	9.3461 ✗	10.5469 ✗	9.6503 ✗	10.3487 ✗
T2c dittography	0.6098	1.9992 ✓	1.3019 ✓	1.6935 ✓	1.4505 ✓	0.9910 ✓
T3a zero replication †	0.0384	0.0403 ✓	0.0391 ✓	0.0518 ✗	0.0473 ✗	0.0551 ✗
T3b slot consistency	0.8683	0.8777 ✓	0.8835 ✓	0.8604 ✓	0.8768 ✓	0.8576 ✓
T3b monotone share	0.5471	0.5828 ✓	0.5943 ✓	0.5541 ✓	0.5804 ✓	0.5532 ✓
T3c final-position entropy	0.6299	0.6535 ✓	0.6534 ✓	0.6593 ✓	0.6540 ✓	0.6665 ✓
T3c initial-position entropy	0.8734	0.8734 ✓	0.8672 ✓	0.8786 ✓	0.8725 ✓	0.8806 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.7808	0.8672 ✓	0.6696 ✓	0.6003 ✓	0.5872 ✓	0.5068 ✓
T6a recency, near repeats †	0.1399	0.1451 ✓	-0.0081 ✗	0.1058 ✓	0.0923 ✗	-0.0130 ✗
T6a recency, exact repeats †	0.2324	0.1925 ✓	0.0050 ✗	0.2737 ✓	0.3201 ✓	-0.0224 ✓
T6b within-page drift †	-0.0082	-0.0016 ✗	-0.0010 ✗	-0.0015 ✗	-0.0017 ✗	-0.0006 ✗
T7b deviant rate †	26.8523	31.6494 ✓	35.6074 ✓	47.0364 ✓	37.9130 ✓	44.4175 ✓
T7b deviant recurrence †	0.0047	0.1868 ✗	0.1458 ✗	0.1214 ✗	0.1520 ✗	0.0906 ✗
T7b slip share	0.2870	0.3955 ✓	0.4417 ✗	0.2873 ✓	0.4148 ✓	0.2741 ✓
E1 line step	0.0591	0.0024 ✓	0.0419 ✓	0.0091 ✓	-0.0004 ✓	0.0070 ✓
E1 paragraph step	-0.0549	-0.0440 ✓	-0.0018 ✓	-0.0164 ✓	-0.0035 ✓	-0.0103 ✓
E1 within-line decay	0.0608	0.0002 ✓	0.0061 ✓	0.0036 ✓	-0.0168 ✓	-0.0154 ✓
E1 within-paragraph decay	0.0447	0.0445 ✓	-0.0080 ✓	0.0290 ✓	0.0036 ✓	-0.0448 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

Table S4. E3 primary direction (fit even, test odd).

Statistic	Voynich	full	no margin rule	no drift	no slips etc.	E2 class
hapax share †	0.7199	0.7227 ✓	0.6800 ✗	0.7173 ✓	0.6246 ✗	0.5970 ✗
type-token ratio †	0.2871	0.3223 ✓	0.2323 ✓	0.3145 ✓	0.2065 ✗	0.1871 ✗
Zipf slope	-0.7499	-0.7776 ✓	-0.7831 ✓	-0.7783 ✓	-0.7772 ✓	-0.7929 ✓
mean word length	5.2924	5.4708 ✓	5.2914 ✓	5.4364 ✓	5.3135 ✓	5.2861 ✓
boundary MI	0.1388	0.1264 ✓	0.1428 ✓	0.1369 ✓	0.1498 ✓	0.1343 ✓
word-pair MI	0.0594	-0.0388 ✗	0.0345 ✓	-0.0615 ✗	-0.0024 ✓	0.0452 ✓
distance-2 word MI	0.0464	0.0168 ✓	0.0260 ✓	0.0161 ✓	0.0392 ✓	0.0413 ✓
line-break word MI	0.0002	0.0047 ✓	0.0031 ✓	0.0027 ✓	0.0173 ✓	0.0042 ✓
line-break boundary MI	0.0078	0.0059 ✓	0.0114 ✓	0.0057 ✓	0.0061 ✓	0.0099 ✓
boundary MI, equal n	0.1480	0.1405 ✓	0.1341 ✓	0.1483 ✓	0.1407 ✓	0.1366 ✓
space predictability	0.7343	0.6962 ✓	0.7540 ✓	0.7009 ✓	0.7816 ✗	0.7896 ✗
lexicon test, real words	0.7455	0.7103 ✓	0.7992 ✓	0.7175 ✓	0.8277 ✓	0.8444 ✗
lexicon test, synthetic words	0.7251	0.6753 ✗	0.7492 ✓	0.6759 ✗	0.7617 ✓	0.7762 ✓
conditional entropy h2	1.9100	1.9981 ✓	1.8345 ✓	1.9888 ✓	1.7647 ✗	1.7471 ✗
conditional entropy h3	1.8075	1.8477 ✓	1.7161 ✓	1.8397 ✓	1.6595 ✗	1.6535 ✗
conditional entropy h4	1.6769	1.7241 ✓	1.6352 ✓	1.7190 ✓	1.6023 ✗	1.5997 ✗
lzma bits per character	2.4581	2.5279 ✓	2.3318 ✓	2.5210 ✓	2.2590 ✓	2.2254 ✗
page MI †	0.3195	0.2646 ✓	0.2450 ✓	0.2530 ✓	0.3493 ✓	0.3620 ✓
T1 line-break ratio	0.3137	-0.1355 ✓	0.0043 ✓	-0.1009 ✓	0.0489 ✓	-0.0051 ✓
T2a final form, margin	0.0789	0.0464 ✓	0.0212 ✗	0.0480 ✓	0.0566 ✓	0.0180 ✗
T2a final form, paragraph end	0.0428	0.0538 ✓	0.0167 ✗	0.0578 ✓	0.0587 ✓	0.0206 ✓
T2b line-fill slack	2.2694	2.6287 ✗	8.0667 ✗	2.6204 ✗	2.7059 ✗	8.1219 ✗
T2c dittography	0.9501	1.3470 ✓	1.6969 ✓	2.1210 ✗	1.3516 ✓	1.7792 ✓
T3a zero replication †	0.0415	0.0574 ✗	0.0547 ✗	0.0621 ✗	0.0313 ✗	0.0298 ✗
T3b slot consistency	0.8684	0.8540 ✓	0.8745 ✓	0.8542 ✓	0.8828 ✓	0.8848 ✓
T3b monotone share	0.5607	0.5277 ✓	0.5785 ✓	0.5353 ✓	0.5936 ✓	0.6047 ✓
T3c final-position entropy	0.6217	0.6402 ✓	0.6147 ✓	0.6358 ✓	0.5886 ✓	0.5839 ✓
T3c initial-position entropy	0.8812	0.8772 ✓	0.8692 ✓	0.8809 ✓	0.8638 ✓	0.8631 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.8021	0.8980 ✓	0.6265 ✓	0.8263 ✓	0.7591 ✓	0.7922 ✓
T6a recency, near repeats †	0.1635	0.1168 ✓	0.1155 ✓	0.1290 ✓	0.0478 ✗	0.1141 ✓
T6a recency, exact repeats †	0.2042	0.2052 ✓	0.0855 ✓	0.1749 ✓	0.0136 ✗	0.1911 ✓
T6b within-page drift †	-0.0084	-0.0019 ✗	-0.0020 ✗	-0.0012 ✗	-0.0016 ✗	-0.0021 ✗
T7b deviant rate †	23.4977	49.0845 ✗	31.9529 ✓	50.6217 ✗	22.9400 ✓	18.7890 ✓
T7b deviant recurrence †	0.0086	0.0681 ✗	0.0487 ✗	0.0642 ✗	0.1837 ✗	0.1836 ✗
T7b slip share	0.2723	0.2960 ✓	0.4209 ✗	0.2797 ✓	0.4257 ✗	0.4163 ✗
E1 line step	0.0655	0.0034 ✓	0.0199 ✓	0.0189 ✓	0.0096 ✓	0.0123 ✓
E1 paragraph step	-0.0244	-0.0318 ✓	0.0055 ✓	-0.0034 ✓	-0.0054 ✓	-0.0012 ✓
E1 within-line decay	0.0692	0.0165 ✓	0.0066 ✓	0.0020 ✓	0.0075 ✓	-0.0005 ✓
E1 within-paragraph decay	0.0817	0.0353 ✓	0.0435 ✓	0.0234 ✓	0.0009 ✓	0.0298 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

Table S5. E6 primary direction (fit even, test odd).

Statistic	Voynich	two-route	no coinage	no repertoire variation	no recency	no boundary rule
hapax share †	0.7199	0.6070 ✗	0.3152 ✗	0.6781 ✗	0.6526 ✗	0.6047 ✗
type-token ratio †	0.2871	0.1336 ✗	0.0640 ✗	0.2890 ✓	0.1877 ✗	0.1327 ✗
Zipf slope	-0.7499	-0.7634 ✓	-0.7907 ✓	-0.7242 ✓	-0.7807 ✓	-0.7716 ✓
mean word length	5.2924	5.0055 ✗	4.8110 ✗	5.5328 ✗	5.1737 ✓	4.9951 ✗
boundary MI	0.1388	0.1256 ✓	0.1238 ✓	0.1287 ✓	0.1393 ✓	0.0005 ✗
word-pair MI	0.0594	-0.0032 ✓	0.0313 ✓	-0.0468 ✗	-0.0151 ✗	-0.0262 ✗
distance-2 word MI	0.0464	0.0503 ✓	0.0746 ✓	0.0125 ✓	0.0450 ✓	0.0458 ✓
line-break word MI	0.0002	0.0322 ✓	0.0536 ✗	0.0182 ✓	0.0272 ✓	0.0407 ✗
line-break boundary MI	0.0078	0.0139 ✓	0.0189 ✓	0.0180 ✓	0.0189 ✓	0.0195 ✓
boundary MI, equal n	0.1480	0.1365 ✓	0.1387 ✓	0.1434 ✓	0.1449 ✓	0.0041 ✓
space predictability	0.7343	0.8110 ✗	0.8484 ✗	0.7342 ✓	0.7830 ✗	0.8276 ✗
lexicon test, real words	0.7455	0.8978 ✗	0.9677 ✗	0.7491 ✓	0.8471 ✗	0.9015 ✗
lexicon test, synthetic words	0.7251	0.8069 ✗	0.8473 ✗	0.6896 ✓	0.7681 ✓	0.8068 ✗
conditional entropy h2	1.9100	1.7184 ✗	1.6104 ✗	1.9271 ✓	1.7833 ✓	1.7405 ✗
conditional entropy h3	1.8075	1.6332 ✗	1.5380 ✗	1.8163 ✓	1.6911 ✗	1.6564 ✗
conditional entropy h4	1.6769	1.5513 ✗	1.4717 ✗	1.7150 ✓	1.6022 ✓	1.5721 ✗
lzma bits per character	2.4581	2.2290 ✗	2.0911 ✗	2.4963 ✓	2.3219 ✓	2.2539 ✓
page MI †	0.3195	0.4378 ✓	0.4760 ✗	0.1916 ✗	0.4043 ✓	0.3943 ✓
T1 line-break ratio	0.3137	-0.0369 ✓	0.2971 ✓	0.0502 ✓	-0.0398 ✓	undef. ✗
T2a final form, margin	0.0789	0.1025 ✓	0.1033 ✓	0.0932 ✓	0.1019 ✓	0.0996 ✓
T2a final form, paragraph end	0.0428	0.0461 ✓	0.0512 ✓	0.0678 ✓	0.0594 ✓	0.0565 ✓
T2b line-fill slack	2.2694	2.7781 ✗	2.6571 ✗	2.7373 ✗	2.5448 ✗	2.7543 ✗
T2c dittography	0.9501	1.1848 ✓	1.2145 ✓	1.1697 ✓	0.9520 ✓	1.2275 ✓
T3a zero replication †	0.0415	0.0353 ✓	0.0285 ✗	0.0464 ✓	0.0377 ✓	0.0330 ✓
T3b slot consistency	0.8684	0.9086 ✗	0.9284 ✗	0.8616 ✓	0.8913 ✓	0.9071 ✗
T3b monotone share	0.5607	0.6645 ✗	0.7103 ✗	0.5279 ✓	0.6164 ✓	0.6506 ✗
T3c final-position entropy	0.6217	0.6148 ✓	0.6017 ✓	0.6468 ✓	0.6263 ✓	0.6149 ✓
T3c initial-position entropy	0.8812	0.8595 ✓	0.8541 ✓	0.8748 ✓	0.8645 ✓	0.8689 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.8021	0.4761 ✓	0.3850 ✗	0.5684 ✓	0.5275 ✓	0.4666 ✓
T6a recency, near repeats †	0.1635	0.0641 ✗	0.0368 ✗	0.0134 ✗	-0.0262 ✗	0.0488 ✗
T6a recency, exact repeats †	0.2042	-0.0005 ✗	-0.0052 ✗	0.1216 ✓	-0.0339 ✗	0.0655 ✗
T6b within-page drift †	-0.0084	-0.0002 ✗	-0.0010 ✗	-0.0008 ✗	-0.0006 ✗	-0.0008 ✗
T7b deviant rate †	23.4977	27.3205 ✓	20.4694 ✓	38.1832 ✓	28.4477 ✓	26.9757 ✓
T7b deviant recurrence †	0.0086	0.0467 ✗	0.1503 ✗	0.0942 ✗	0.0602 ✗	0.0499 ✗
T7b slip share	0.2723	0.4817 ✗	0.8110 ✗	0.2929 ✓	0.3348 ✓	0.4602 ✗
E1 line step	0.0655	0.0330 ✓	0.0103 ✓	0.0055 ✓	0.0130 ✓	0.0381 ✓
E1 paragraph step	-0.0244	0.0341 ✓	-0.0033 ✓	0.0250 ✓	-0.0117 ✓	0.0118 ✓
E1 within-line decay	0.0692	0.0018 ✓	0.0074 ✓	-0.0079 ✓	0.0063 ✓	0.0070 ✓
E1 within-paragraph decay	0.0817	0.0209 ✓	0.0033 ✓	0.0348 ✓	-0.0171 ✗	0.0092 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

S4. Drift and burstiness

Table S6. E5: page clustering index B by frequency band and gradient G with quire-jackknife 95% intervals.

Corpus	G	B 2–4	B 5–19	B 20–99	B ≥ 100
Voynich	+ 0.24 (− 0.10, 0.59)	2.02	1.83	1.58	1.36
Voynich (Takahashi)	+ 0.19 (− 0.28, 0.66)	1.91	1.87	1.58	1.34
Latin	+ 0.74 (0.57, 0.92)	2.95	2.29	1.40	1.07
Italian	+ 0.78 (0.47, 1.09)	2.59	1.56	1.19	1.02
German	+ 1.06 (0.81, 1.30)	4.89	2.96	1.70	1.13
English	+ 1.08 (0.93, 1.22)	5.11	2.61	1.74	1.22
VB-run	+ 0.66 (0.49, 0.82)	2.23	1.48	1.15	1.03
NAIB-run	+ 0.07 (− 0.33, 0.46)	1.09	1.04	1.02	1.00
LDRIFT	− 0.19 (− 0.52, 0.14)	2.22	3.02	2.67	2.12
REP	− 0.60 (− 1.14, − 0.05)	1.53	2.06	2.77	2.40
HGR2	+ 0.97 (0.49, 1.44)	3.68	1.70	1.40	1.31
MK-sec	+ 0.03 (− 0.28, 0.34)	1.10	1.15	1.07	1.03

Table S7. E4: drift slopes of line similarity over line distance 2–8 with page-bootstrap 95% intervals.

Corpus	L (lexical)	S (sublexical)
Voynich	− 0.0142 (− 0.0275, − 0.0022)	− 0.0052 (− 0.0085, − 0.0023)
Latin	− 0.0115 (− 0.0257, 0.0040)	− 0.0028 (− 0.0045, − 0.0011)
Italian	− 0.0147 (− 0.0255, − 0.0026)	− 0.0121 (− 0.0153, − 0.0088)
German	− 0.0252 (− 0.0402, − 0.0116)	− 0.0043 (− 0.0070, − 0.0015)
English	− 0.0128 (− 0.0206, − 0.0042)	+ 0.0003 (− 0.0031, 0.0040)
VB-run	− 0.0092 (− 0.0190, − 0.0001)	+ 0.0002 (− 0.0020, 0.0022)
NAIB-run	+ 0.0076 (− 0.0075, 0.0242)	− 0.0015 (− 0.0043, 0.0012)
GDRIFT	− 0.0517 (− 0.0660, − 0.0385)	− 0.0148 (− 0.0176, − 0.0112)
LDRIFT	− 0.0398 (− 0.0498, − 0.0291)	+ 0.0007 (− 0.0023, 0.0033)
MK-sec	− 0.0042 (− 0.0151, 0.0064)	− 0.0007 (− 0.0030, 0.0016)

S5. Definitions of the 40 phenotype statistics

Earlier battery (companion manuscript):

- hapax share and type-token ratio;
- Zipf slope of the 200 most frequent types;
- mean word length;
- boundary MI: excess MI between the last glyph of a word and the first glyph of the next inside lines, against a within-line permutation null;
- word-pair MI and distance-2 word MI;
- line-break MI and line-break boundary MI, with the boundary MI at equal n;
- space predictability: information gain about word boundaries from the flanking glyph pair;
- lexicon test: share of held-out real words and of glyph-trigram samples attested in the training vocabulary;

- conditional glyph entropies h2–h4;
- lzma bits per character;
- page MI between page identity and word.

Mechanism battery: T1 page-conditional line-break ratio; T2a final-form excess (margin-bound, paragraph-final); T2b slack ratio (stars section); T2c dittography ratio; T3a zero replication (250-token chunks); T3b slot consistency and monotone share; T3c positional entropy (final, initial); T4 any periodicity peak; T5 step/smooth prediction-error ratio; T6a near and exact recency indices; T6b slope of line similarity over line distance 1–8; T7b deviant rate, recurrence share and slip share.

E1: graded line step, paragraph step, graded within-line decay, within-paragraph decay.

Tolerance: $|G - V| \leq 2\sqrt{SE_V^2 + SD_G^2} + 0.01|V|$. SE_V is a leave-one-quiere-out jackknife over the test half; SD_G is taken over three generator seeds. T4 passes if at most one seed shows a peak.

S6. Implementation notes

- **Page-conditional T1 estimator.** Per page, the excess plug-in MI over 200 within-page re-pairings, for the break pairs and for consecutive blocks of within-line pairs of the same size, averaged over pages with weights equal to the number of break pairs. The pooled estimators failed a known-truth simulation in which each page of Latin received a different letter substitution; this estimator returned identical values with and without substitution.
- **E1 expectations.** Position classes: offset from the line start (1 or ≥ 2), offset from the line end (1 or ≥ 2), paragraph-final line or not. The binary expectation is the exact mean over pairs of distinct eligible tokens of the page in the same class pair.
- **E5 null.** For each type, the expected same-page pairs are $\sum_g C(n_{tg}, 2) \cdot \sum_{p \in g} N_p(N_p - 1) / (N_g(N_g - 1))$, with groups $g = \text{hand} \times \text{section}$.
- **E7 coder.** 32-bit Witten–Neal–Cleary arithmetic coding over integer transition counts. The end-of-line suppression below 60% of the target length and the stop at 140% are deterministic and replicated by the receiver.